

Exploring Generative AI: CycleGAN, Transformer NMT, and Vision Transformers

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Abstract. This report investigates three advanced generative AI tasks: unpaired image translation using CycleGAN, English-Urdu neural machine translation with Transformer and mBART, and a comparative analysis of CNN and Vision Transformer models on CIFAR-10. Each section details the motivation, methodology, experiments, and findings, providing unique insights into the effectiveness of these models.

1 Introduction

Generative AI has revolutionized both vision and language domains. This report presents implementations and evaluations of CycleGAN for image translation, Transformer-based models for neural machine translation, and Vision Transformers for image classification. The focus is on practical challenges, model architectures, and empirical results.

2 CycleGAN for Unpaired Image Translation

2.1 Motivation

Translating between faces and sketches without paired data is a challenging task. CycleGAN enables learning such mappings using cycle consistency and adversarial training.

2.2 Methodology

The CycleGAN framework utilizes two generators (ResNet-9) and two PatchGAN discriminators. The loss function combines adversarial, cycle consistency, and identity losses. The Person Face Sketches dataset is used, with data augmentation and checkpointing for robust training.

2.3 Experiments and Results

The model is trained for 200 epochs with batch size 1. Visual inspection shows realistic translations, though some artifacts remain. Cycle consistency loss helps preserve facial structure, but fine details can be lost.

3 English-Urdu Neural Machine Translation

3.1 Motivation

Machine translation between English and Urdu is complex due to linguistic differences and limited resources. Transformer-based models and mBART offer promising solutions.

3.2 Methodology

A custom Transformer is implemented with multi-head attention and positional encoding. mBART is fine-tuned on the parallel corpus. BLEU score is used for evaluation. Data preprocessing includes normalization and tokenization.

3.3 Experiments and Results

The custom Transformer achieves moderate BLEU scores, while mBART performs better, especially in fluency and rare word translation. Pre-training and transfer learning are key advantages of mBART.

4 CNN vs Vision Transformer on CIFAR-10

4.1 Motivation

Comparing CNNs and Vision Transformers highlights the evolution of image classification models. ViTs leverage self-attention for global context, while CNNs are efficient for local features.

4.2 Methodology

A custom CNN with residual connections and a Vision Transformer (ViT) are trained on CIFAR-10. Data augmentation and early stopping are applied. The ViT is implemented from scratch and compared with a pre-trained version.

4.3 Experiments and Results

ViT outperforms the CNN in accuracy, particularly on complex classes. CNNs converge faster but are limited in modeling long-range dependencies. Data augmentation benefits both models.

5 Conclusion

This study demonstrates the strengths of CycleGAN for unpaired image translation, the superiority of mBART in low-resource translation, and the effectiveness of Vision Transformers for image classification. Future work could explore larger datasets and hybrid architectures.

References

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